

Competency

- To identify the geometric figures of similar shapes and solve the related problems

Learning Outcomes

- To indicate the condition of similarity on polygons (triangles and polygons) and solve the problems related problems

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define similarity on polygons - To tell the conditions of similarity of triangles - To recall the Pythagoras theorem
2.	Understanding (U)	- To find the sides of similar triangles or polygons - To show the similarity of given triangles
3.	Application (A)	- To prove the theorems based on similarity of triangles
4.	High Ability (HA)	- To prove the Pythagoras theorem by the virtue of the similarity of triangles - To explore the required relations based similar triangles

Required Teaching Materials/ Resources

Various photos of same shapes, card boards of different sizes but same shapes, leaves of same tree, models of enlargement and reduction etc

Pre-knowledge: Congruent triangles, similar triangles

Teaching Activities

- For warm-up, show the pair of square/circular sheet of paper, Nepali flags, leaves, photos etc and discuss upon the properties so observed
- Divide the students into 4 groups and do the following activities
 - Provide a pair of similar triangles to group-A
 - Provide a pair of similar quadrilaterals to group-B
 - Provide a pair of similar pentagons to group-C
 - Provide a pair of similar hexagons to group-D

Then tell them to discuss in own group, measure the corresponding sides and angles, their relations and present the own result in the class.
- Also, ask the following questions
 - Are the corresponding sides of similar triangles equal?
 - Are the corresponding sides of similar triangles proportional?
 - Are the corresponding angles of similar triangles equal?
 - Are the corresponding angles of similar polygons equal?
 - Are the corresponding sides and perimeter of similar polygons equal?
 - Are the corresponding sides and diagonals of similar polygons equal?

- Discuss on the conditions of similarity of triangles
- Solve the problems from the exercise with discussion.
- Focus on project work

Solution of selected problems from Vedanta Excel in Mathematics

1. In the given figure, $\triangle ABC \sim \triangle ABD$, $\angle BAD = \angle ACB$, find the length of BD.

Solution:

Here, $\triangle ABC \sim \triangle ABD$

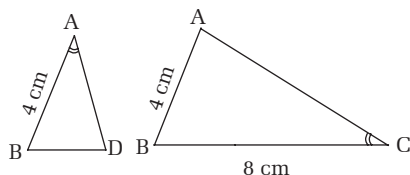
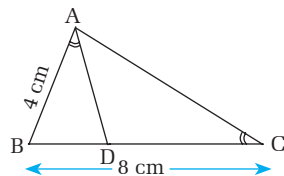
$$\therefore \frac{AB}{BD} = \frac{AB}{BC}$$

[Corresponding sides of similar triangle are proportional]

$$\text{or, } \frac{4 \text{ cm}}{BD} = \frac{8 \text{ cm}}{4 \text{ cm}}$$

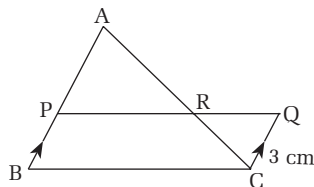
$$\therefore BD = 2 \text{ cm}$$

Hence, the length of BD is 2 cm.



2. In the adjoining figure, $AB \parallel QC$, $PR = 2RQ$ and $QC = 3 \text{ cm}$. Find the length of AP with the suitable reasons.

Solution:



	Statements		Reasons
1.	In $\triangle APR$ and $\triangle QCR$	1.	
i.	$\angle APR = \angle RQC$	i.	$AB \parallel QC$ and alternate angles
ii.	$\angle PAR = \angle RCQ$	ii.	Same as (i)
iii.	$\angle ARP = \angle RQC$	iii.	Vertically opposite angles are equal
2.	$\triangle APR \sim \triangle QCR$	2.	By A.A.A. axiom
3.	$\frac{AP}{QC} = \frac{PR}{RQ}$ or, $\frac{AP}{3 \text{ cm}} = \frac{2 \text{ RQ}}{RQ} \therefore AP = 6 \text{ cm}$	3.	Corresponding sides of similar triangle are proportional

3. In a given right angled triangle ABC right angled at B, $BD \perp AC$ and $\angle CBD = \angle BAC$. Prove that:

(i) $\triangle ABC \sim \triangle BCD$

(ii) $\triangle BCD \sim \triangle ABD$

(iii) $BC^2 = AC \cdot CD$ (iv) $BD^2 = AD \cdot CD$ (v) $\frac{BC^2}{BD^2} = \frac{AC}{AD}$

Solution:

Given: $\triangle ABC$ is a right angled triangle in which $\angle ABC = 90^\circ$,
 $BD \perp AC$ and $\angle CBD = \angle BAC$

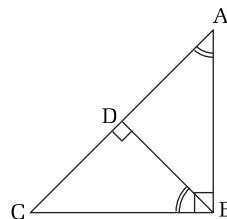
To prove: (i) $\triangle ABC \sim \triangle BCD$

(ii) $\triangle BCD \sim \triangle ABD$

(iii) $BC^2 = AC \cdot CD$

(iv) $BD^2 = AD \cdot CD$

(v) $\frac{BC^2}{BD^2} = \frac{AC}{AD}$



Proof

	Statements		Reasons
1.	In $\triangle ABC$ and $\triangle BCD$	1.	
i.	$\angle ABC = \angle BCD$	i.	Both are right angles
ii.	$\angle ACB = \angle BCD$	ii.	Common angles
2.	$\triangle ABC \sim \triangle BCD$	2.	By A.A fact
3.	$\frac{AB}{BD} = \frac{BC}{CD} = \frac{AC}{BC}$ $\therefore BC^2 = AC \cdot CD$	3.	Corresponding sides of similar triangle are proportion From 2 nd and 3 rd ration
4.	In $\triangle BCD$ and $\triangle ABD$	4.	
i.	$\angle BDC = \angle ADB$	i.	Both are right angles
ii.	$\angle DBC = \angle BAD$	ii.	Given
5.	$\triangle BCD \sim \triangle ABD$	5.	By A.A. fact
6.	$\frac{BC}{AB} = \frac{BD}{AD} = \frac{CD}{BD}$ $\therefore BD^2 = AD \cdot CD$	6.	Corresponding sides of similar triangle are proportion From 2 nd and 3 rd ration
7.	$\frac{BC^2}{BD^2} = \frac{AC \cdot CD}{AD \cdot CD} = \frac{AC}{AD}$	7.	From (3) and (6)

Proved

4. In the given figure, $\angle ADE = \angle ACB$ and $\angle DAC = \angle BAE$.
Prove that, $AD \cdot BC = AC \cdot DE$.

Solution:

Given: $\angle ADE = \angle ACB$ and $\angle DAC = \angle BAE$.

To prove: $AD \cdot BC = AC \cdot DE$

Proof

	Statements		Reasons
1.	$\angle DAC = \angle BAE$	1.	Given
2.	$\angle DAE = \angle BAC$	2.	Adding $\angle EAC$ on both sides of (1)
3.	In $\triangle ADE$ and $\triangle ABC$	3.	
i.	$\angle ADE = \angle ACB$	i.	Given
ii.	$\angle DAE = \angle BAC$	ii.	From (2)
4.	$\triangle ADE \sim \triangle ABC$	4.	By A.A axiom
5.	$\frac{AE}{AB} = \frac{DE}{BC} = \frac{AD}{AC}$	5.	Being corresponding sides of similar triangles
6.	$AD \cdot BC = AC \cdot DE$	6.	From (5), taking last two ration

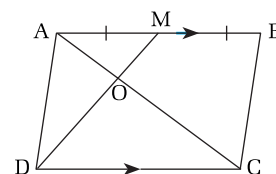
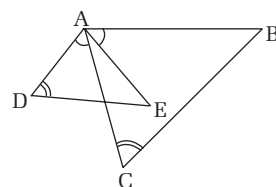
Proved

5. In the given figure, $AB = DC$, $AB \parallel DC$ and M is the mid - point of AB . Prove that:
(i) $\triangle AOM \sim \triangle COD$ (ii) $CO = 2AO$

Solution:

Given: $AB = DC$, $AB \parallel DC$ and M is the mid - point of AB .

To prove: (i) $\triangle AOM \sim \triangle COD$ (ii) $CO = 2AO$



Proof

	Statements		Reasons
1.	In $\triangle AOM$ and $\triangle COD$	1.	
i.	$\angle OAM = \angle OCD$	i.	$AB \parallel DC$ and alternate angles
ii.	$\angle AOM = \angle COD$	ii.	Vertically opposite angles
2.	$\triangle AOM \sim \triangle COD$	2.	By A.A. axiom
3.	$\frac{AO}{CO} = \frac{OM}{OD} = \frac{AM}{CD}$	3.	Corresponding sides of similar triangle
4.	$AB = 2AM$ and $AB = CD$	4.	Given
5.	$\frac{AO}{CO} = \frac{AM}{2AM} \therefore CO = 2AO$		

Proved

6. In the given figure, $AB \parallel MN \parallel DC$. If $AB = x$, $DC = y$ and $MN = z$.

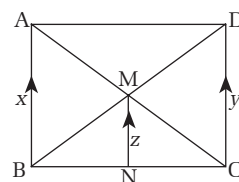
Prove that $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$.

Solution:

Given: $AB \parallel MN \parallel DC$, $AB = x$, $DC = y$ and $MN = z$.

To prove: $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

Proof

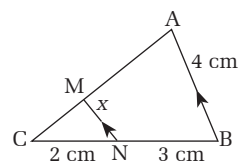


	Statements		Reasons
1.	In $\triangle ABC$ and $\triangle MNC$	1.	
i.	$\angle ABC = \angle MNC$	i.	$AB \parallel MN$, corresponding angles
ii.	$\angle BAC = \angle NMC$	ii.	Same as (i)
iii.	$\angle ACB = \angle MCN$	iii.	Common angles
2.	$\triangle ABC \sim \triangle MNC$	2.	By A.A.A. axiom
3.	$\frac{AB}{MN} = \frac{BC}{NC}$ or, $\frac{x}{z} = \frac{BC}{NC} \therefore NC = \frac{z}{x} BC$		Corresponding sides of similar triangles
4.	In $\triangle BCD$ and $\triangle BNM$	4.	
i.	$\angle BCD = \angle BNM$	i.	$MN \parallel DC$, corresponding angles
ii.	$\angle BDC = \angle BMN$	ii.	Same as 4(i)
iii.	$\angle DBC = \angle MBN$	iii.	Common angles
5.	$\triangle BCD \sim \triangle BNM$	5.	By A.A.A. axiom
6.	$\frac{DC}{MN} = \frac{BC}{BN}$ or, $\frac{y}{z} = \frac{BC}{BN} \therefore BN = \frac{z}{y} BC$	6.	Corresponding sides of similar triangles
7.	$NC + BN = zBC \left(\frac{1}{x} + \frac{1}{y} \right)$	7.	Adding (3) and (6)
8.	$BC = zBC \left(\frac{1}{x} + \frac{1}{y} \right) \therefore \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$	8.	$NC + BN = BC$

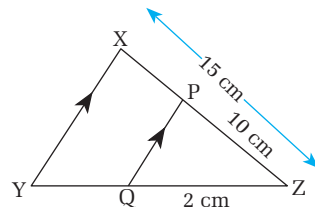
Proved

Extra Questions

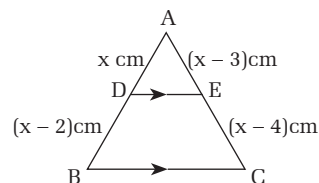
1. In the given figure, $\triangle ABC \sim \triangle MNC$, find the value of x .
[Ans: 1.6 cm]



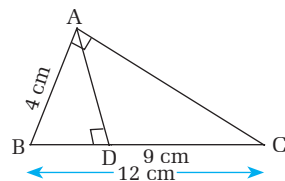
2. In the adjoining figure, $XZ = 15$ cm, $PZ = 10$ cm, $QZ = 6$ cm and $XY \parallel PQ$. Find the length of YQ .
[Ans: 3 cm]



3. In the figure given alongside $\triangle ABC \sim \triangle ADC$. Find the value of x .
[Ans: 6]

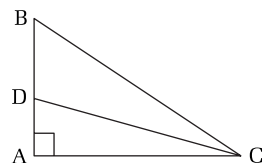


4. In the given figure, $\angle BAC = 90^\circ$, $AD \perp BC$, $CD = 9$ cm and $BC = 12$ cm, find the length of AB .
[Ans: 6 cm]



B. Pythagorean Theorem

1. In the adjoining right-angled triangle ABC , D is any point on AB . Prove that $AB^2 - AD^2 = BC^2 - CD^2$.

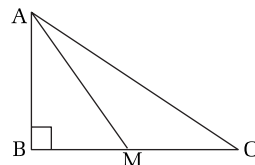


Solution:

	Statements		Reasons
1.	$AB^2 + AC^2 = BC^2$ $\therefore AC^2 = BC^2 - AB^2$	1.	Using Pythagoras theorem in $\triangle ABC$
2.	$AD^2 + AC^2 = CD^2$ $\therefore AC^2 = CD^2 - AD^2$	2.	Using Pythagoras theorem in $\triangle ADC$
3.	$BC^2 - AB^2 = CD^2 - AD^2$ $\therefore BC^2 - CD^2 = AB^2 - AD^2$	3.	From (1) and (2)

Proved

2. In the given right-angled triangle ABC , M is the mid-point of BC . Prove that $AC^2 = AM^2 + 3CM^2$.



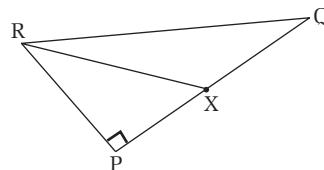
Solution:

	Statements		Reasons
1.	$AB^2 + BM^2 = AM^2$	1.	By Pythagoras theorem

2.	$AB^2 + CM^2 = AM^2$ $\therefore AB^2 = AM^2 - CM^2$	2.	$BM = CM$ and from (1)
3.	$AB^2 + BC^2 = AC^2$	3.	By Pythagoras theorem
4.	$AB^2 + (2\text{cm})^2 = AC^2$	4.	From (3), $BC = 2\text{ cm}$
5.	$AM^2 - CM^2 + 4\text{ cm}^2 = AC^2$ $\therefore AC^2 = AM^2 + 3\text{ cm}^2$	5.	From (2) and (4)

Proved

3. In the given figure, X is the mid - point of PQ . Prove that $PQ^2 = (RX^2 - PR^2)$

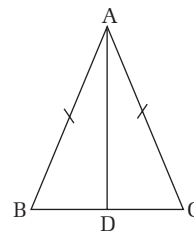


Solution:

	Statements		Reasons
1.	$PQ = 2PX$ $\therefore PQ^2 = (2PX)^2 = 4PX^2$	1.	X being the mid - point of PQ
2.	$PX^2 = RX^2 - PR^2$	2.	In $\triangle PRX$, by Pythagoras theorem
3.	$PQ^2 = 4(RX^2 - PR^2)$	3.	From (1) and (2)

Proved

4. ABC is an isosceles triangle in which $AB = AC = 2BC$. If AD is an altitude of the triangle. Prove that $4AD^2 = 15BC^2$.

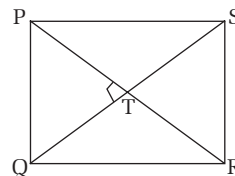


Solution:

	Statements		Reasons
1.	$BD = CD = \frac{1}{2} BC$	1.	Altitude of isosceles triangle bisects its base
2.	$AD^2 + BD^2 = AB^2$ or, $AD^2 + (\frac{1}{2}BC)^2 = (2BC)^2$ or, $4AD^2 + BC^2 = 16BC^2$ $\therefore 4AD^2 = 15BC^2$	2.	In $\triangle ABD$, by Pythagoras theorem $BD = \frac{1}{2} BC$ and $AB = 2BC$

Proved

5. In the given figure, diagonals of a quadrilateral PQRS are intersected at T at right angle. Prove that: $PQ^2 + SR^2 = QR^2 + PS^2$



Solution:

Given: The diagonals of the quadrilateral PQRS intersect at T at a right angle.

To prove: $PQ^2 + SR^2 = QR^2 + PS^2$

Proof

	Statements		Reasons
1.	In ΔPQT , $QT^2 = PQ^2 - PT^2$ In ΔQTR , $QT^2 = QR^2 - TR^2$ Thus, $PQ^2 - PT^2 = QR^2 - TR^2$ $\therefore PQ^2 = QR^2 + PT^2 - TR^2$	1.	By using Pythagoras theorem
2.	In ΔPST , $ST^2 = PS^2 - PT^2$ In ΔSTR , $ST^2 = SR^2 - TR^2$ Thus, $PS^2 - PT^2 = SR^2 - TR^2$ $\therefore SR^2 = PS^2 - PT^2 + TR^2$	2.	Same as (1)
3.	$PQ^2 + SR^2 = QR^2 + PT^2 - TR^2 + PS^2 - PT^2 + TR^2$ $\therefore PQ^2 + SR^2 = QR^2 + PS^2$	3.	From (1) and (2)

Proved

6. In the figure alongside, O is any point interior to the rectangle.

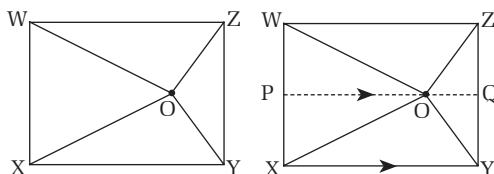
Prove that $OX^2 - OY^2 = OW^2 - OZ^2$.

Solution:

Given: O is any point interior to the rectangle $WXYZ$.

To prove: $OX^2 - OY^2 = OW^2 - OZ^2$

Construction: $PQ \parallel XY$ is drawn through O where P is on WX and Q is on YZ



Proof

	Statements		Reasons
1.	$OP \perp XW$ and $OQ \perp YZ$	1.	$PQ \parallel XY$ and $\angle X = \angle Y = 90^\circ$
2.	$OP^2 = OW^2 - PW^2$ and $OP^2 = OX^2 - PX^2$ or, $OW^2 - PW^2 = OX^2 - PX^2$ $\therefore OX^2 = OW^2 + PX^2 - PW^2$	2.	In ΔPOW and ΔPOX using Pythagoras theorem
3.	$OQ^2 = OZ^2 - QZ^2$ and $OQ^2 = OY^2 - QY^2$ or, $OZ^2 - QZ^2 = OY^2 - QY^2$ $\therefore OY^2 = OZ^2 - QZ^2 + QY^2$	3.	In ΔQOZ and ΔQOY using Pythagoras theorem
4.	$OX^2 - OY^2 = OW^2 + PX^2 - PW^2 - (OZ^2 - QZ^2 + QY^2)$ $= OW^2 - OZ^2 + PX^2 - PW^2 + QZ^2 - QY^2$	4.	Subtracting (3) from (2)
5.	$OX^2 - OY^2 = OW^2 - OZ^2 + QY^2 - PW^2 + PW^2 - QY^2$ $\therefore OX^2 - OY^2 = OW^2 - OZ^2$	5.	$PX = QY$, $QZ = PW$

Proved